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Date

Vid-68 : Vector Subspaces & Examples

Let V be a non-empty set of elements (n-tuples, matrices, polynomials, etc.).
Let each element of V be called a vector.
Define two operations ' ' :
→ vector addition (+)
$u, v \in V \Rightarrow u + v \in V$
→ Scalar multiplication (·)
For scalars α, $\alpha \cdot u \in V$

V is called a vector space if for $a, b, c \in V$ and any scalars α, β, following properties are satisfied:
Refer notes pdf. (8ms) (10 axioms make up vector spaces
left & right distribution are different)

Vector space V is denoted as $(V, +, \cdot)$

$C[a,b]$ represents set of all continuous functions defined on $[a,b]$

Let $f, g \in C[a,b]$
Then, $(f+g)(x) = f(x) + g(x)$
$f(x) + g(x) = (f+g)(x) \in C[a,b]$
Also for scalar α, $(\alpha f)(x) = \alpha f(x) \in C[a,b]$
∴, $C[a,b]$ is a vector space.
(From notes Pdf bro)

Example Let V be a set of real valued continuous functions defined on $[0,1]$ such that $f(\frac{2}{3}) = 2$. Check whether V is a vector subspace or not?

Sol Given $V = \{f \mid f \in C[0,1], f(\frac{2}{3}) = 2\}$
Let $f, g \in V$
Then $(f+g)(\frac{2}{3}) = 4$
∴, $f + g \notin V$
∴, V is not a vector space

V_n is a set of all n-tuples
i.e. V_3 is equivalent to R^3

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~~Subspace~~

A non-empty ~~set~~ W is ~~set to be~~ said to be a subspace of V

Subspace

A non-empty subset W of vector space V is said to be a subspace of V if W is vector space under the same operations of addition & multiplication as V.

(i.e. if $u, v \in W$ and $\forall \alpha, \beta \in R \Rightarrow (\alpha u + \beta v) \in W$)

To prove subset W is a subspace of vector space V_3, show that :

For $u, v \in W$ and $\alpha \in R$

$u + v \in W$

$\alpha u \in W$

$0 \in W$ (zero element must be in W)

Example Let V be a vector space in R^3. Check whether

i) $W = \{(a,b,c) \mid a \leq 0\}$ is a subspace

Sol Let $u = (a,b,c) \in W$

Take $\alpha = -4 \in R$

Then

$-4u = (-4a, -4b, -4c)$

Since $-4 < 0$ and $a < 0$

$\Rightarrow -4a > 0$

$\therefore -4u \notin W$

$\therefore$ W is not subspace.

ii) $W = \{(a,b,c) \mid c$ is integer$\}$

Ans Not a subspace.

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Example Check whether set of all symmetric matrices in $M_{n\times n}$ are subspace of $M_{n\times n}$.

Sol: Let $W = \{A \in M_{n\times n} \mid A^T = A\}$

Let $A, B \in W$

Then $A^T = A$ & $B^T = B$

→ And $(A+B)^T = A^T + B^T = A+B$

$\therefore (A+B) \in W$ →

→ Also, $(\alpha A)^T = \alpha A^T = \alpha A$ → Hence W is a subspace of $M_{n\times n}$

$\therefore (\alpha A) \in W$ →

Example Check if $W = \{(x,y,z) \mid \frac{y}{x} = \sqrt{2}\}$ is a subspace of R^3

Sol: Not a subspace as for $\alpha = 0$ and $(x,y,z) \in W$ we have:

$\alpha \cdot (x,y,z) = (0,0,0)$ hence $\frac{0}{0} \neq \sqrt{2}$

Vid-70, Linear Span, Combination | Smallest Subspace

(Set of scalars)

Linear Combination

Let V be a vector space over field F. If $v_1, v_2 \ldots v_n \in V$ then element $v \in V$ written as:

$$v = \sum_{i=1} \alpha_i v_i$$

(Where $\alpha_i \in F$) is called linear combination of vector.

Linear span

If S is a non-empty subset of vector space V(F), then the set of all linear combination of the elements of S is called the linear span of S.

It is denoted by [S] or L(S)

Result $L(S)$ is the smallest subspace containing S, where S is a subset of $V(F)$.

i.e. $L(S) = \langle S \rangle$

Example In vector space $V_3(R)$, consider the vectors $v_1 = (1,2,1)$, $v_2 = (3,1,5)$, $v_3 = (3,-4,7)$. Show that the subspaces spanned by $S = \{v_1, v_2\}$ & $T = \{v_1, v_2, v_3\}$ are the same

Sol: "Need to show $L(S) = L(T)$"

"To show that show (i) $L(S) \subseteq L(T)$ and $L(T) \subseteq L(S)$"

→ To show $L(S) \subseteq L(T)$, show that every element of $L(S)$ can be written $L(S)$ can be written as linear combination of $L(T)$. (eg)

→ To show $L(T) \subseteq L(S)$, show that every element of $L(T)$ can be written as linear combination of $L(S)$

$v_1 = 1v_1 + 0v_2$

$v_2 = 0v_1 + 1v_2$

$v_3 = ?v_1 + ?v_2$

Find & done.

Unit-II: Linear Independence / LI & LD

If V be a vector space over field F, then vectors $v_1, v_2, v_3 \dots v_n \in V$ are said to be linearly dependent over F, if $\exists \alpha_i \in F$ such that,

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \dots + \alpha_n v_n = 0$$

where at least one of $\alpha_i \neq 0$

They are said to be linearly independent over F if $\exists \alpha_i \in F$ such that

$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \dots + \alpha_n v_n = 0$

Implies each $\alpha_i = 0$.

Remarks: LD & LI depend on field F.

Eg example $V_1 = 1$ and $V_2 = i$

V_1 and V_2 are linearly ~~independent~~ dependent over complex field, but linearly independent over real field.

4 methods to check for linear dependence in ↑video ~~notes~~, but only one in notes. So only going use system of equations solving

To check for L.I & L.D. of functions

Two ~~differentiable~~ functions f_1 & f_2 are linearly dependent if there are non-zero constants α_1 & α_2 such that $\alpha_1 f_1 + \alpha_2 f_2 = 0$.

For three or more functions, define Wronskian as

$$\begin{vmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{vmatrix} = \begin{cases} = 0 \text{ ; functions are L.D. } \forall x \in R \\ \neq 0 \text{ ; L.I. for some } x_0 \in R \end{cases}$$

We say S spans V if every element in V can be written as linear combination of elements in S. i.e. $[S] = V$

Vid-72: Part-1 Basis and dimension of vector space

Let $V(F)$ be a vector space. A subset S of V is called a basis of V if and only if

→ S is linearly independent set

→ $L(S) = V$ or $[S] = V$ (i.e. S generates or spans V)

$V = P_n$ - set of all polynomials with degree $\leq n$

$V = V_n = R^n$

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Dimension of vector space

If vector space & V has basis S that contains finite no. of elements then space is called finitely dimensioned or finitely generated. (otherwise infinite dimensional)

The no. of elements in basis of V is called dimension of vector space V.
It is denoted by dim(V)

→ $V = P_n$ will have basis $\{1, x, x^2 \dots, x^n\}$ and has $n+1$ dimensions

→ $V = V_n = R^n$ will have basis $\{(1,0,0\dots), (0,1,0\dots), \dots, (0,0\dots,1)\}$ and will have n dimensions.

i.e. $\dim(P_n) = n+1$

$\dim(V_n) = n$ (n-dimensional space)

Example let $W = \{(x_1, x_2, x_3) \in R^3 \mid x_1 - 3x_2 + 3x_3 = 0\}$ be a subspace of vector space R^3. Find the basis and dimension of W.

Sol^n $W = \{(x_1, x_2, x_3) \mid x_1 = 3x_2 - 3x_3\}$

$W = \{(3x_2 - 3x_3, x_2, x_3) \forall x_2 \in R^3\}$

$W = \{x_2(3,1,0) + x_3(-3,0,1), x_2, x_3 \in R\}$

∴ any element of W can be represented as a linear combination of ~~any~~ two. (3,1,0) & (-3,0,1)

Also both vectors are L.I.

∴ basis is $\{(3,1,0), (-3,0,1)\}$ and $\dim(W) = 2$

Theorem: In a n-dimensional vector space V, any set of n linearly independent vectors is a basis.

→ In this case we only need to check that the given set is L.I. or not for the basis.

The condition [S] = V is not required to check/proof.

(Refer notes pdf it has good info & lecture)

Vid-78: Eigen Values and Eigen Vectors

Let $A=[a_{ij}]$ be a square matrix of order $n \times n$ over field F, then

$$|A-\lambda I_n| = \begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \\ \cdots & \cdots & \cdots \end{vmatrix}$$

I_n identity matrix of order n

$|A-\lambda I_n|$ is called characteristic polynomial of A.

Eigen Values

Let $A=(a_{ij})$ be square matrix of order n.

The roots of characteristic equation

$$|A-\lambda I|=0$$

are defined as characteristic roots / latent roots / eigen values of matrix A.

→ The roots & eigenvalues may be (real & complex) and (distinct & repeated)

→ Spectrum of matrix A is the set of distinct eigen values of A.

→ The largest absolute value of λ_i's is called the spectral radius of A

Eigen Vector

A non-zero vector X is called the eigen vector corresponding to eigen value λ, if it satisfies homogeneous equation.

$$AX=\lambda X$$

i.e. $(A-\lambda I_n)X=0$

Example

Sol

→ If eigenvalues λ_i's are distinct we get n linearly independent eigenvectors.

→ If λ_i's has some repeated eigenvalues, it may or may not be possible to get L.I. eigenvectors.

→ If X is eigenvector of matrix A corresponding to eigenvalue λ, then so is kX, $k \neq 0$
Thus eigenvector corresponding to eigenvalue is not unique.

Trivial solution $X=0$ of $AX=\lambda X$ is of no interest. Need to find non-trivial solutions.

Geometrically, matrix A stretches vector X by λ.

Rule:-
→ Sum of eigenvalues = trace of matrix (Sum of diagonal)
→ Product of eigenvalues = $|A|$

→ Determinant of matrix formed by eigenvectors is nonzero.

Example: Find eigenvalue and eigenvector of matrix:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$$

Solution: Solving,

$$AX = \lambda X$$

$$(A - \lambda I_3)X = 0$$

$$|A - \lambda I_3| = 0$$

For this to have non-trivial solution (i.e. $X \neq 0$), $|A - \lambda I_3| = 0$